

“Hand” DES Example

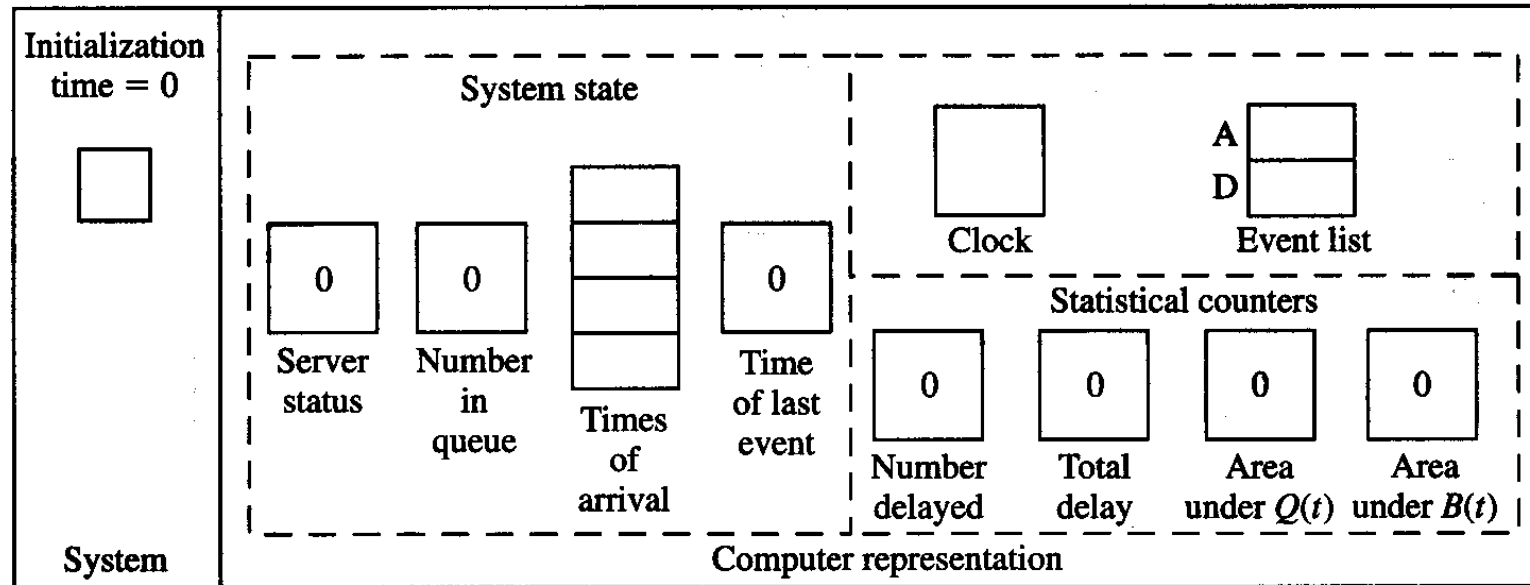
- Given (for now) inter-arrival times (all times are in minutes):
 - 0.4, 1.2, 0.5, 1.7, 0.2, 1.6, 0.2, 1.4, 1.9, ...
- Given service times:
 - 2.0, 0.7, 0.2, 1.1, 3.7, 0.6, ...
- $n = 6$ delays in queue desired
- “Hand” simulation:
 - Display system, state variables, clock, event list, statistical counters ... all after execution of each event
 - Use above lists of inter-arrival, service times to “drive” simulation
 - Stop when number of delays hits $n = 6$, compute output performance measures

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- Quantities to be estimated
 - *Expected average delay in queue* (excluding service time) of the n customers completing their delays
 - Why “expected?”
 - *Expected average number of customers in queue* (excluding any in service)
 - A continuous-time average
 - Area under $Q(t)$ = queue length at time t , divided by $T(n)$ = time simulation ends
 - *Expected utilization (proportion of time busy) of the server*
 - Another continuous-time average
 - Area under $B(t)$ = server-busy function (1 if busy, 0 if idle at time t), divided by $T(n)$
 - Many others are possible (maxima, minima, time or number in system, proportions, quantiles, variances, etc.)
- Important: *Discrete-time vs. continuous-time* statistics

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Status shown is *after* all changes have been made in each case ...



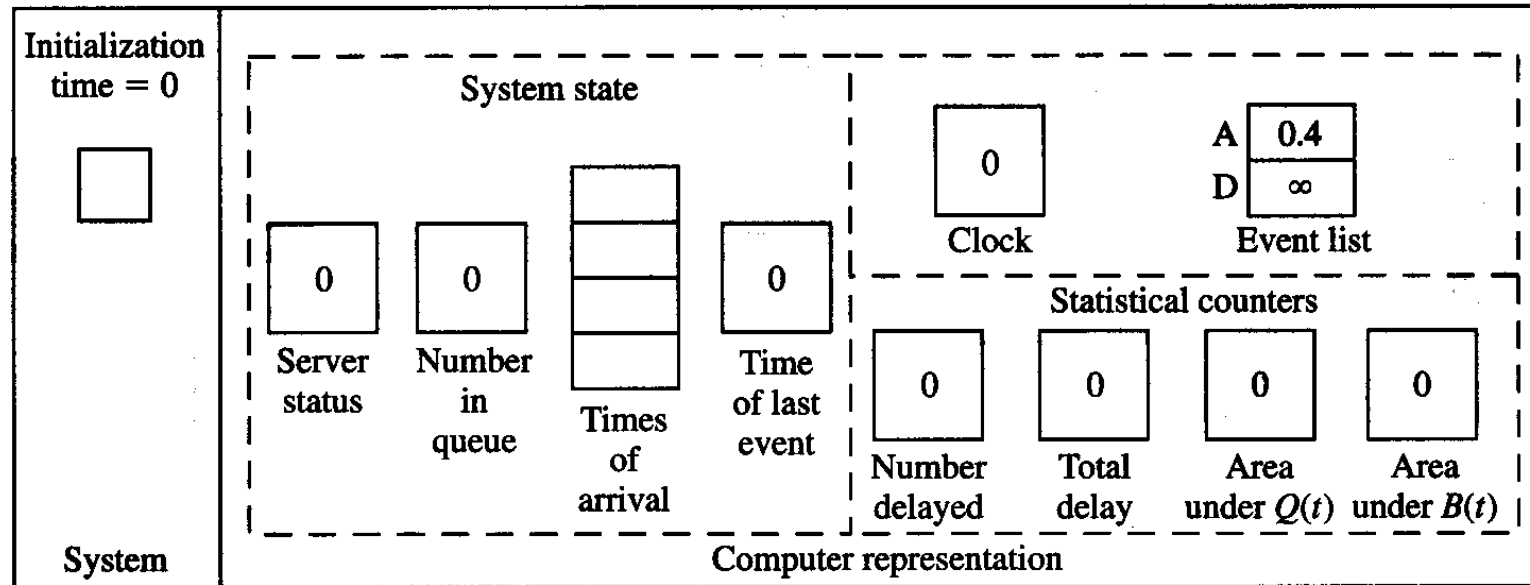
Interarrival times: 0.4, 1.2, 0.5, 1.7, 0.2, 1.6, 0.2, 1.4, 1.9, ...
 Service times: 2.0, 0.7, 0.2, 1.1, 3.7, 0.6, ...

When will future events be scheduled?

- **Departure Event** -> when a new customer is admitted to the server
- **Arrival Event** -> when an arrival occurs (according to the next interarrival)

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Status shown is *after* all changes have been made in each case ...

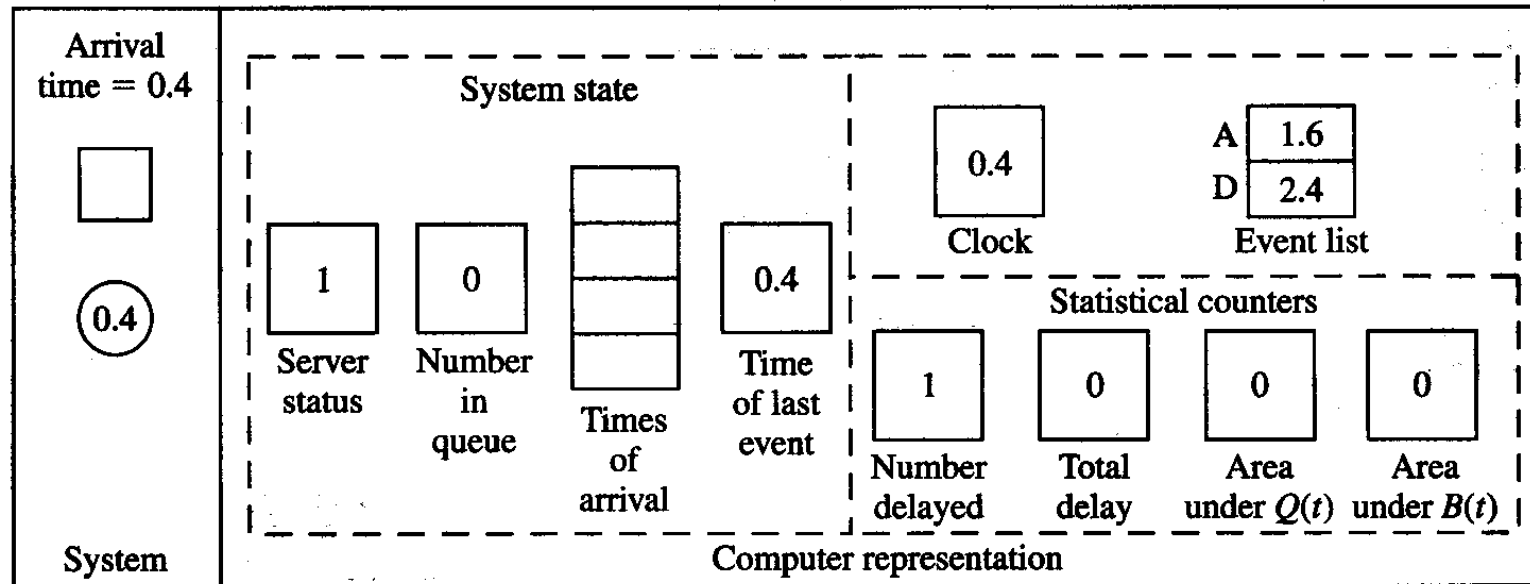


Interarrival times: ~~0.4~~, 1.2, 0.5, 1.7, 0.2, 1.6, 0.2, 1.4, 1.9, ...
 Service times: 2.0, 0.7, 0.2, 1.1, 3.7, 0.6, ...

For each iteration:

- The **next event (event with smallest timestamp)** is selected
- The simulation time is advanced
- The event is processed:
 - State variables/statistical variables are updated
 - New future events are possibly generated

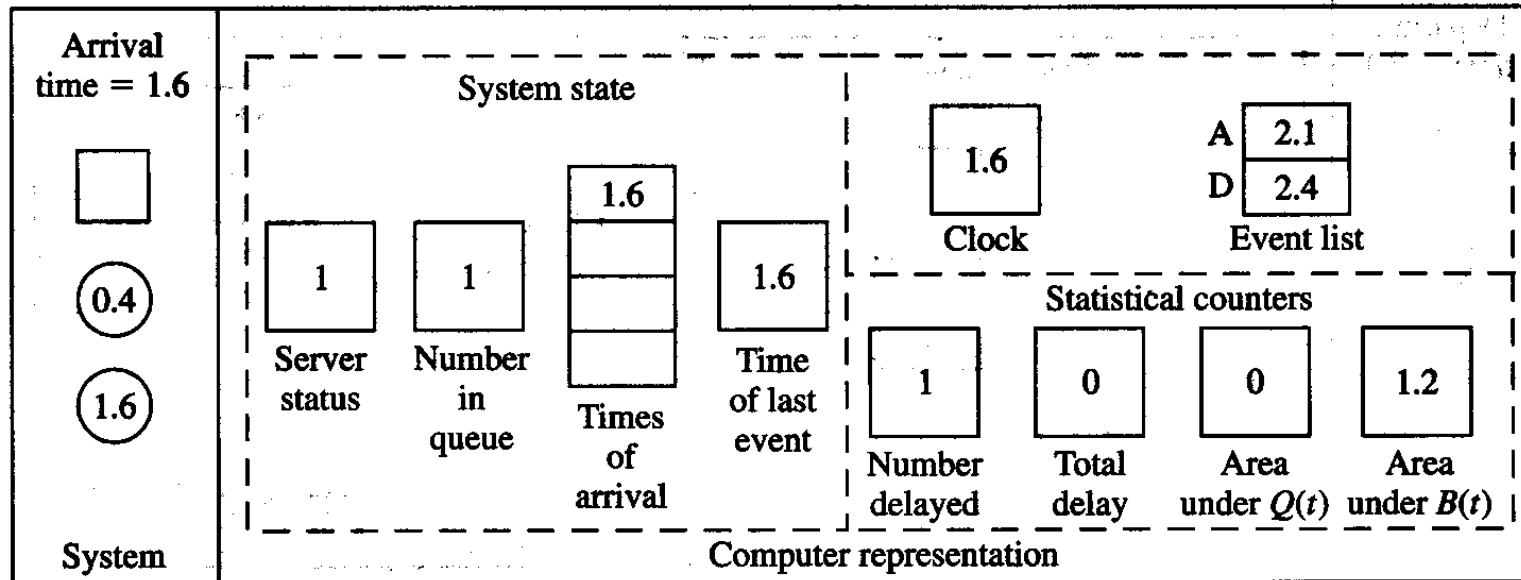
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Interarrival times: ~~0.4~~, ~~1.2~~, 0.5, 1.7, 0.2, 1.6, 0.2, 1.4, 1.9, ...

Service times: ~~2.0~~, 0.7, 0.2, 1.1, 3.7, 0.6, ...

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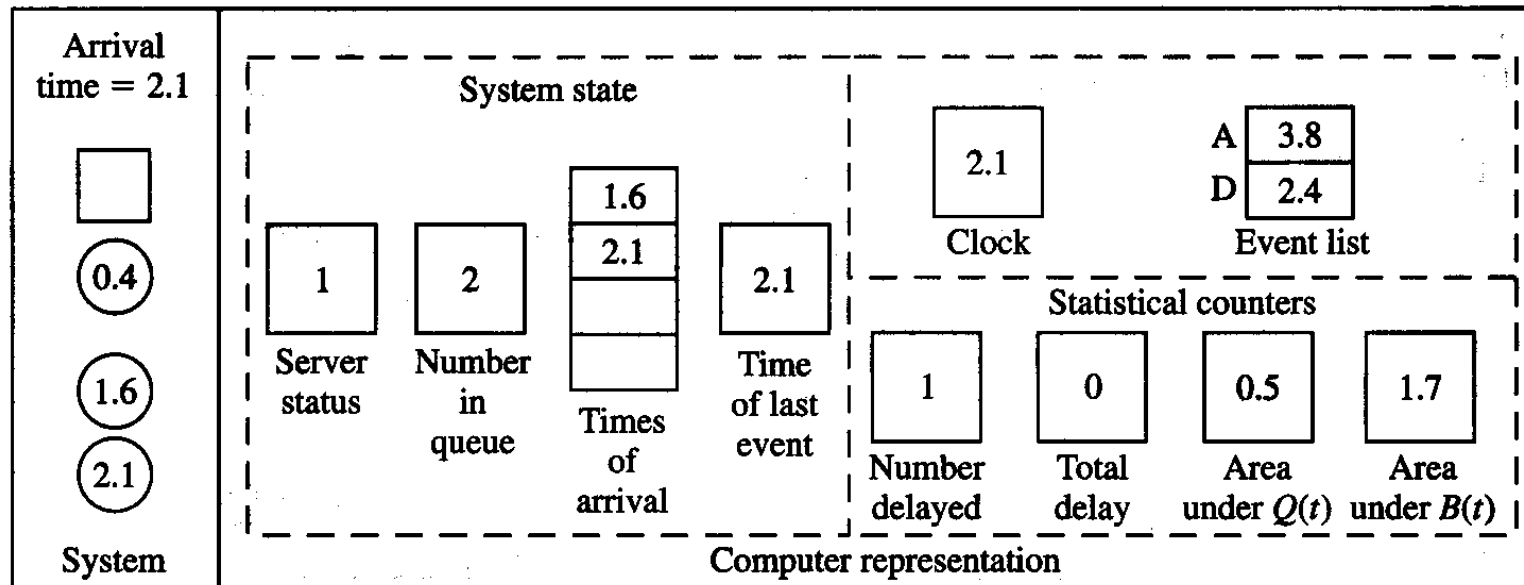
Interarrival times: ~~0.4~~, ~~1.2~~, ~~0.5~~, 1.7, 0.2, 1.6, 0.2, 1.4, 1.9, ...

Service times: ~~2.0~~, 0.7, 0.2, 1.1, 3.7, 0.6, ...

How to compute the Area under $B(t)$ for each time step:

- $\Delta T = \text{Clock} - \text{timestep_of_previousEvent}$
 - $1.6 - 0.4 = 1.2$
- If server is busy in ΔT
 - $\text{Area under } B(t) += \Delta T$ (else do nothing)

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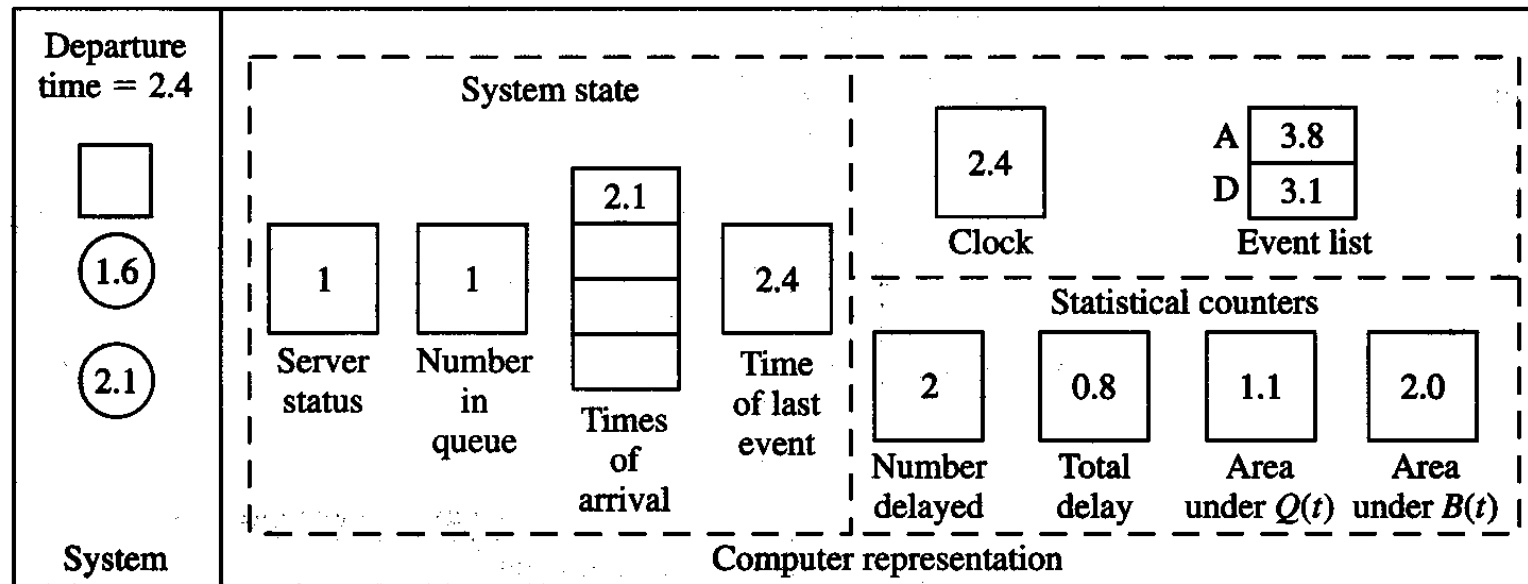
Interarrival times: ~~0.4~~, ~~1.2~~, ~~0.5~~, ~~1.7~~, 0.2, 1.6, 0.2, 1.4, 1.9, ...

Service times: ~~2.0~~, 0.7, 0.2, 1.1, 3.7, 0.6, ...

How to compute the Area under $Q(t)$ for each time step:

- $\Delta T = \text{Clock} - \text{timestep_of_previousEvent}$
 - $2.1 - 1.6 = 0.5$
- Let N = number of customers in queue in ΔT
 - $\text{Area under } Q(t) += \Delta T * N$

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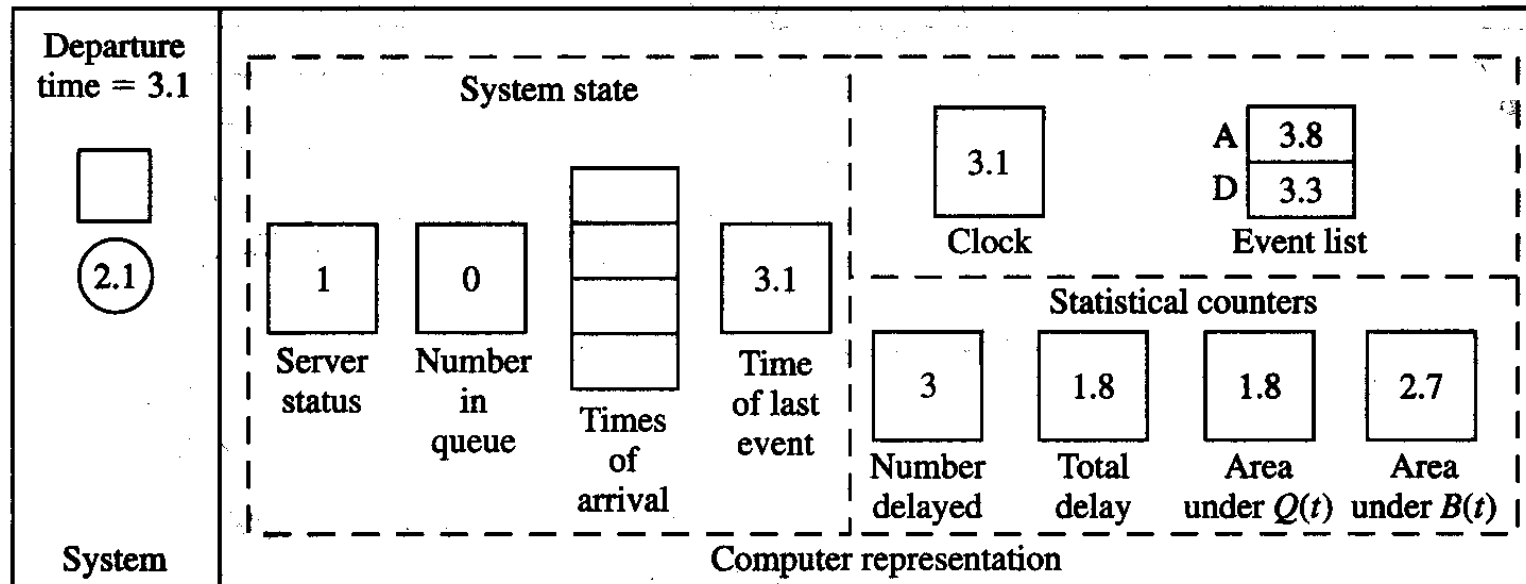
Interarrival times: ~~0.4~~, ~~1.2~~, ~~0.5~~, ~~1.7~~, 0.2, 1.6, 0.2, 1.4, 1.9, ...

Service times: ~~2.0~~, ~~0.7~~, 0.2, 1.1, 3.7, 0.6, ...

- $\Delta T = 2.4 - 2.1 = 0.3$
- Area Under $B(t) += 0.3 \rightarrow 1.7 + 0.3$
- Area under $Q(t) += 0.3 * 2 \rightarrow 0.5 + 0.6 = 1.1$

- Total delay += delay of current job admitted to the server
- = current time - job time of arrival = $2.4 - 1.6 = 0.8$

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Interarrival times: ~~0.4~~, ~~1.2~~, ~~0.5~~, ~~1.7~~, 0.2, 1.6, 0.2, 1.4, 1.9, ...

Service times: ~~2.0~~, ~~0.7~~, ~~0.2~~, 1.1, 3.7, 0.6, ...

- $\Delta T = 3,1 - 2,4 = 0,7$
- Area Under $B(t)$ += 0,7 -> $2,0 + 0,7 = 2,7$
- Area under $Q(t)$ += $1 * 0,7$ -> $1,1 + 0,7 = 1,8$
- Total delay += $3,1 - 2,1$ -> $0,8 + (3,1 - 2,1) = 1,8$